

Multi-Target Ocean Wave Forecasting Using Deep Learning

This repository accompanies our work on multi-target ocean wave forecasting using deep learning models.

Publication

Title: Multi-Target Ocean Wave Forecasting Using Deep Learning: Comparative Analysis of Recurrent and Transformer-Based Architectures

IEEE Xplore: <https://ieeexplore.ieee.org/document/11580576>

Overview

This work compares three deep learning architectures for forecasting six oceanographic parameters from historical wave data:

- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)
- Temporal Fusion Transformer (TFT)

The models are evaluated using RMSE, MAE, MAPE, and R^2 . Experimental results show that the Temporal Fusion Transformer (TFT) achieves the best overall performance by effectively capturing long-term temporal dependencies.

Applications

- Coastal safety
- Maritime navigation
- Port operations
- Marine logistics
- Ocean monitoring

Citation

If you use this work in your research, please cite the IEEE publication:

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